

Public Economics Practicum 4

Discussion Questions Rosen:

	8th edition	9th edition
Chapter 3	8	10
Chapter 8	5	5
Chapter 11	3	3
Chapter 12	7, 8	7, 8
Chapter 13	1, 4, 7	1, 4, 7

Rosen 8.5 and 13.7 will be discussed in the tutorial.

Additional Exercises:

- 4.1 Suppose there are two individuals, André and Bernard, who must split 400 euro. André's utility of income equals

$$U_A = \alpha \ln I_A$$

and for Bernard, it equals

$$U_B = \beta \ln I_B$$

where I_A and I_B are the amounts of income André and Bernard receive, respectively, and $\alpha > 0$ and $\beta > 0$.

- (a) What is the optimal distribution of income if the social welfare function is additive ($W = U_A + U_B$)? Explain the effects of α and β .
- (b) Suppose $\alpha = \beta = 1$. What is the optimal distribution of income if the social welfare function is given by $W = \omega_A U_A + \omega_B U_B$? Explain the effects of ω_A and ω_B .

- 4.2 Dunya and Nikki have identical utility functions:

$$U_i = \sqrt{I_i}$$

where I_i is monthly income. Suppose that Dunya works, and earns 2000 euro per month, whereas Nikki does not have a job and, hence, does not earn any money. Suppose that the government can freely redistribute income, and that the social welfare function is additive.

- (a) What is the optimal distribution of income?

Now suppose that the government incurs costs in redistributing income. In particular, the cost of giving Y to Nikki is equal to $0.1Y$.

- (b) How much income is optimally collected from Dunya, and how much does Nikki receive (round off at whole euro's)?

4.3 A country has the following welfare program for helping the poor. Each person has the right to a benefit of G euro per month, but for each euro someone earns, the benefit is reduced with t euro. Suppose that $G = 600$ and $t = 0.4$. Consider Betty, who can work for an hourly wage of 10 euro. Assume that a month has 30 days (such that total number of hours $T = 720$).

- (a) What is the break-even point of the welfare program, in terms of earnings and hours worked?
- (b) What is the mathematical expression of the budget constraint under the welfare program? Sketch Betty's budget constraint with and without the existence of the welfare program.

Now suppose that Betty's utility of income (in euro's) Y and leisure (in hours) L equals

$$U = Y^{\frac{1}{6}} L^{\frac{5}{6}}$$

- (c) What is Betty's optimal amount of leisure if there would be no welfare program?
- (d) What is the Betty's optimal amount of leisure under the welfare program? How large is the benefit she receives? What is her utility?

Now suppose that the welfare program would be such that Betty receives a benefit equal to the exact amount she received under d), but now independent of her own earnings.

- (e) What would be Betty's optimal amount of leisure? Why is it different from c) and d)?
- (f) Compare Betty's utility under d) and e). How does this relate to the trade-off between efficiency and distribution?

4.4 Consider three individuals, Anna, Bert, and Dirk. Anna earns $Y > 0$ euro and Bert also earns Y euro. Dirk earns 0 euro. The government engages in redistribution, by transferring T euro from Anna to Dirk and T euro from Bert to Dirk (so that Dirk receives in total $2T$). Anna, Bert, and Dirk maximise their

own utility and consume all of their disposable income (after redistribution). Anna's utility is given by $U_A = \sqrt{C_A}$, Bert's utility is given by $U_B = \sqrt{C_B}$, and Dirk's utility is given by $U_D = w\sqrt{C_D}$, where C_A , C_B , and C_D are the levels of consumption of Anna, Bert, and Dirk, respectively, in euro's. The Social Welfare function is given by $SW = U_A + U_B + U_D$. Throughout, we assume that parameter $w > 0$.

- (a) Derive the transfer T that maximises social welfare.

The optimal transfer T (as derived under a) increases in w .

- (b) Explain why the optimal transfer T increases in w . (note: do not provide a mathematical description, but an economic explanation. You can also answer this question if you did not solve question a).

Now suppose that the government does not engage in redistribution, $T = 0$. Furthermore, assume now that Anna and Bert both care about Dirk. In particular, Anna's utility equals $U_A = \sqrt{C_A} + \frac{1}{2}U_D$ and Bert's utility equals $U_B = \sqrt{C_B} + \frac{1}{2}U_D$. Both Anna and Bert consider whether to give some of their income to Dirk. Anna and Bert decide independently of each other how much of their income to give to Dirk. Assume from now on that $w = 1$.

- (c) Derive how much Anna gives to Dirk in the Nash-equilibrium, and how much Bert gives to Dirk in the Nash equilibrium.
- (d) From the perspective of Anna and Bert, the outcome under c) is not efficient. Explain in words why the outcome under c) is not efficient from the perspective of Anna and Bert. Ignore the perspective of Dirk. Note: use the correct type of market failure in your answer and explain why it leads to an inefficient outcome in this situation.

4.5 (this is an old exam exercise with propositions)

Aïsha and Nora are sisters. Nora is relatively rich. Her income $Y_N = 2500$ euro. Aïsha is relatively poor, her income is only $Y_A = 400$ euro. Aïsha's utility can be given by

$$U_A = \sqrt{C_A}$$

where C_A is her level of consumption in euro's. Nora's utility function can be given by

$$U_N = \sqrt{C_N} + \alpha U_A$$

where C_N is Nora's level of consumption in euro's and α is a parameter. We assume that $0 \leq \alpha < 1$. Nora considers giving Aïsha some amount of money $G \geq 0$. After this gift, both sisters spend their disposable income on consumption, implying that $C_N = Y_N - G$ and $C_A = Y_A + G$.

- (a) If $\alpha > 0.4$, then $G > 0$.
- (b) If $\alpha = 0.5$, then a gift of $G = 500$ euro leads to a pareto-improvement compared to $G = 0$.

Now suppose that the government introduces a public poverty alleviation programme. Through this programme, Aïsha receives a benefit of $B = 200$ euro, implying that $C_A = Y_A + B + G$.

- (c) If $\alpha = 0.5$, then $G = 20$ euro.
- (d) If $\alpha > 0.4$, the introduction of the public poverty alleviation programme leads to crowding-out of private poverty alleviation.

Answers exercises week 4

Rosen 3.8 / 3.10 / 3.10

Social welfare is maximized when Mark's marginal utility of income is equal to Judy's marginal utility of income. Taking the derivative of Mark's utility function to find his marginal utility function yields $MU_M = \frac{50}{\sqrt{I_M}}$ and taking the derivative of Judy's utility function yields $MU_J = \frac{100}{\sqrt{I_J}}$. If we set MU_M equal to MU_J , the condition for maximization becomes $I_J = 4I_M$ and, since the fixed amount of income is \$300, this means that Mark should have \$60 and Judy should have \$240 if the goal is to maximize social welfare $= U_M + U_J$.

Rosen 8.5

a) Bill is willing to pay 25 cents to save 5 minutes, so he values time at 5 cents per minute. The subway saves him 10 minutes per trip, or 50 cents. The value of 10 trips per year is \$5. The cost of each trip is 40 cents, or \$4 per year. The annual net benefit to Bill is therefore \$1. The present value of the benefits $= \$5/0.25 = \20 ; the present value of the costs is $\$4/0.25 = \16 .

Note: the present value of an income stream of x per year that starts next year and lasts forever equals $\frac{x}{r}$, where r is the interest rate.

b) Total benefits $= \$20 \times 55,000 = \$1,100,000$. Total costs $= \$16 \times 55,000 = \$880,000$. Net benefits $= \$220,000$.

c) Costs $= \$1.25 \times 55,000 = \$68,750$. Benefits $= (\$62,500/1.25) + (\$62,500/1.25^2) = \$90,000$. Net benefit $= \$21,250$.

d) The subway project has a higher present value. If a dollar to the "poor" is valued the same as a dollar to the "middle class," choose the subway project.

e) Let λ be the distributional weight of a poor person relative to a rich person. The two projects are equivalent in terms of social welfare if

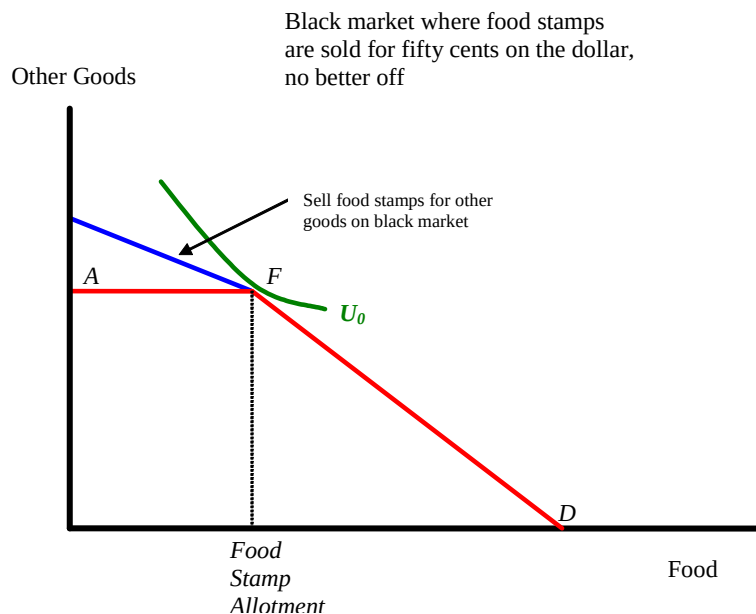
$$220,000 = -68,750 + \lambda \left(\frac{62,500}{1.25} + \frac{62,500}{[1.25]^2} \right)$$

which can be solved for $\lambda = 3.21$. In other words, this distribution weight means that \$1 of income to a poor person must be viewed as more important than \$3.21 to the middle class for the legal services to be implemented rather than the subway project.

Rosen 12.7 / 12.7 / 12.8

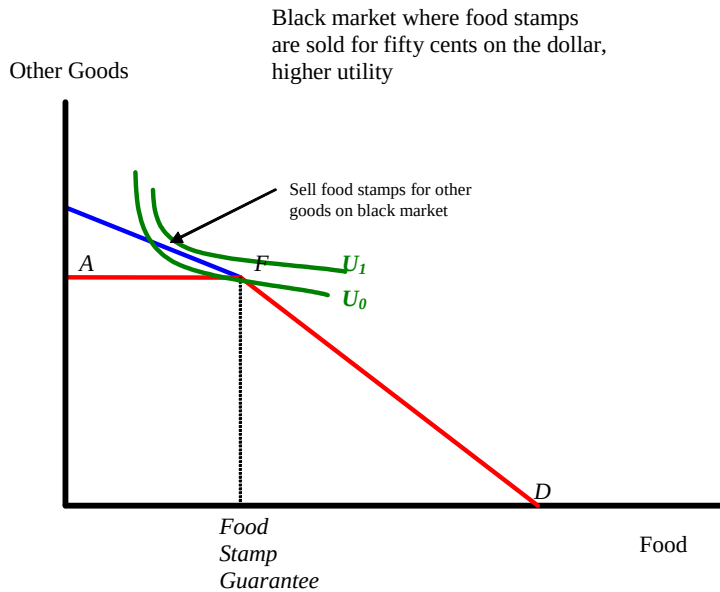
Initially the price of food was \$2 and the price of other goods was \$1. The black market for food stamps changes the price of food sold to \$1. In Figure 12.3 of the textbook, as one moves to the "northwest" from point F, the segment will now have a slope (in absolute value) of 1 rather than 2. The black market may make the individual better off if the best point on her budget constraint AFD was initially at

the corner solution of point F, and the black market certainly does not make her worse off. It is important to note that the black market does not always make the recipient better off. If the (absolute value) of the marginal rate of substitution (MRS) were between 1 and 2, the indifference curve would not “cut” into the new part of the budget constraint with the black market.



Rosen 12.7A

If the MRS were less than (or equal to) 1 in absolute value, the person would be made better off and would reduce food consumption by selling the food stamps on the black market.



Rosen 12.7B

Rosen 12.8 / 12.8 / 12.9

Pareto efficient redistribution is a reallocation of income that makes some (or all) people better off while no-one is made worse off. This is impossible if people only care about their own income. However, with these two consumers, Marsha's utility increases as Sherry's utility increases. Thus, it may be possible to reallocate income from Marsha to Sherry and raise both of their utility. With Sherry's initial utility function of $U_S = 100\sqrt{Y_S}$, her utility with $Y_S = 100$ equals $U_S = 1,000$. With Marsha's initial utility function of $U_M = 100\sqrt{Y_M} + 0.8U_S$, her utility with $Y_M = 100$ of income is $U_M = 1000 + 0.8 \cdot 1000 = 1800$. If the social welfare function is additive, then initial welfare is $W = U_S + U_M = 2,800$.

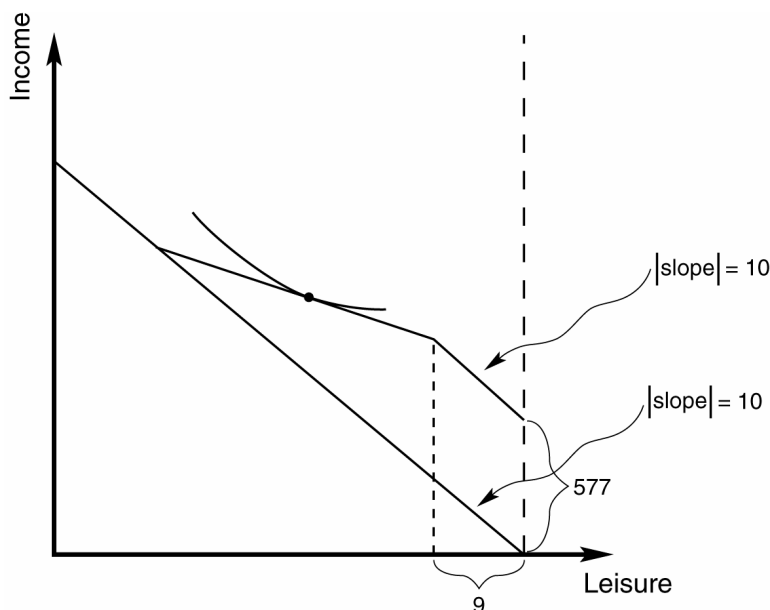
If \$36 is reallocated from Marsha to Sherry, then Sherry's utility becomes $U_S = 1,166.19$. Marsha's utility becomes $U_M = 800 + 932.952 = 1,732.95$. In this case, Sherry's utility increases from 1,000 to 1,166.19, while Marsha's utility falls from 1,800 to 1,732.95. Social welfare increases with this redistribution, going from 2,800 to 2,899.14. Thus, this redistribution increases social welfare, but it is not Pareto efficient redistribution.

Rosen 13.1

Note that the figure below shows the correct shape of the budget constraint, but the numbers themselves are outdated.

a) With a wage rate of \$10 per hour, Elizabeth earns \$100. Because the deduction

in California is \$225, none of her earnings are counted against the \$645 welfare benefit. Thus, her total income is \$745 (=\$100+\$645).



Rosen 13.1

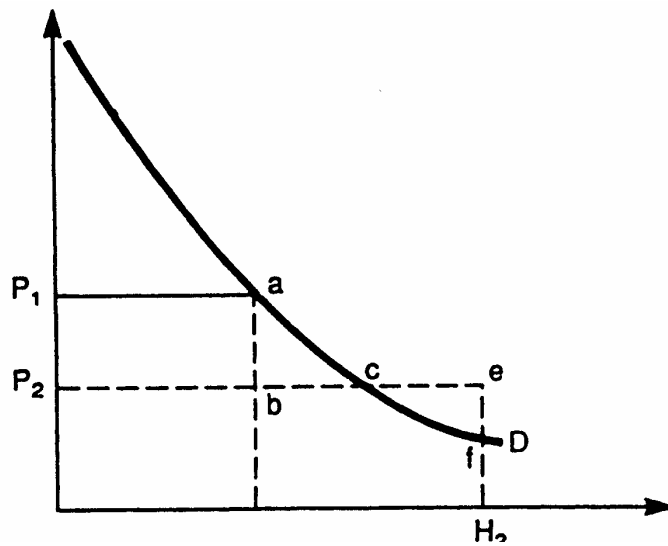
b) The actual welfare benefits collected by a person equals $B = G - t(Earnings - D)$, where B = actual benefits, G = welfare grant, t = tax rate on earned income, and D = standard deduction. Thus, $(Earnings - D)$ is the net earnings that are taxed away in the form of reduced benefits. When benefits equal zero ($B = 0$), the expression becomes $0 = G - t(Earnings - D)$, which can be rewritten to: $Earnings = G/t + D$. This is known as the “breakeven formula.” In the California context here, the expression becomes $Earnings = \$645/0.5 + 225$, or $Earnings = \$1,515$. With a wage rate of \$10 per hour, this corresponds to 151.5 hours of work per month.

c) See Figure. The number 577 should be replaced with 645 and 9 hours should be replaced with 22.5 hours.

d) The diagram above shows one possibility. In this case, Elizabeth is both working and on welfare, but she collects a reduced welfare benefit.

Rosen 13.4

a) He participates in the public housing program as long as $P_1 P_2 c a > c e f$.



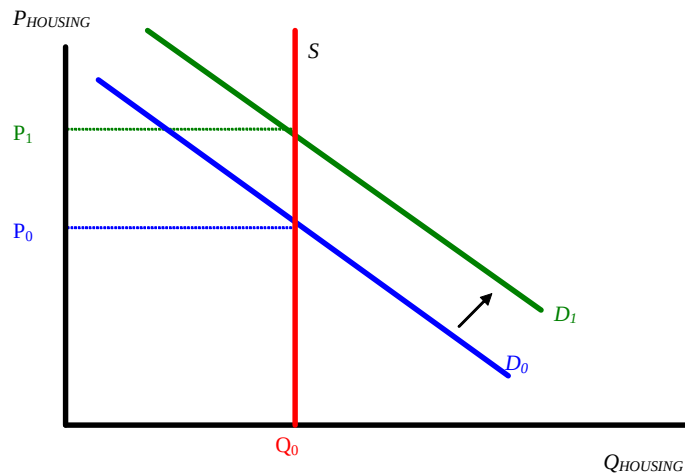
Rosen 13.4

Rosen 13.7

In all cases, the demand curve for housing slopes downward.

a) If the price of low income housing gets bid up but there is no increase in the stock of housing, then the supply curve is perfectly inelastic (vertical).

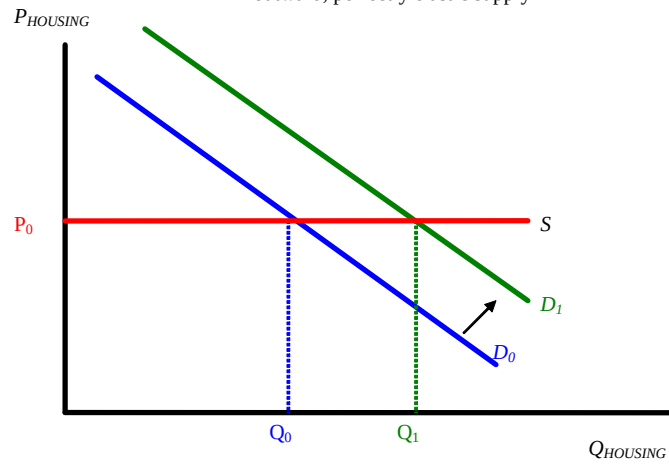
FIGURE 8.7a – Demand curve shifts outward, perfectly inelastic supply



Rosen 13.7A

b) If there is no increase in the price of housing, but there is an increase in the stock of housing, then the supply curve is perfectly elastic (horizontal).

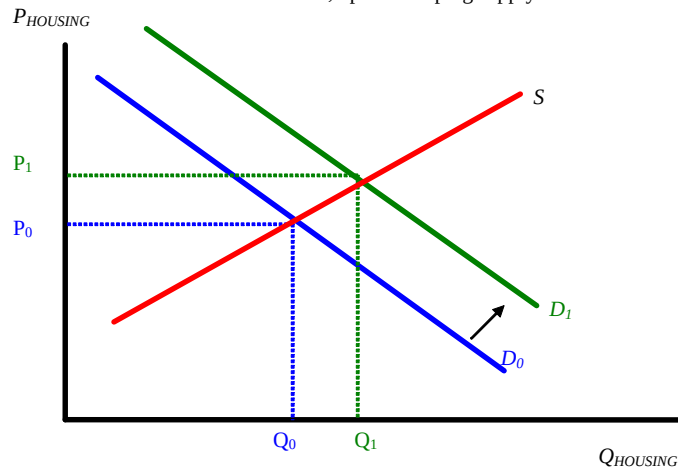
FIGURE 8.7b – Demand curve shifts outward, perfectly elastic supply



Rosen 13.7B

- c) If there is an increase in both the price and quantity of housing, then the supply curve slopes upward.

FIGURE 8.7c – Demand curve shifts outward, upward sloping supply curve



Rosen 13.7C

Rosen 11.3

Use the basic formula for balance in a pay-as-you-go social security system:

$$t = \frac{N_b}{N_w} \frac{B}{w}$$

Call 1990 year 1 and 2050 year 2. Then

$$\begin{aligned} t_1 &= 0.267 \left(\frac{B_1}{w_1} \right) \\ t_2 &= 0.458 \left(\frac{B_2}{w_2} \right) \end{aligned}$$

It follows that to keep $\frac{B_1}{w_1} = \frac{B_2}{w_2}$ we require $\frac{t_2}{t_1} = \frac{0.458}{0.267} = 1.72$. That is, tax rates would have to increase by 72 percent. Similarly, to keep the initial tax rate constant, we would require $\frac{B_2}{w_2} / \frac{B_1}{w_1} = \frac{.267}{.458} = 0.58$. Relative to wages, benefits would have to fall almost by half.

Rosen 8.5

a) Bill is willing to pay 25 cents to save 5 minutes, so he values time at 5 cents per minute. The subway saves him 10 minutes per trip, or 50 cents. The value of 10 trips per year is \$5. The cost of each trip is 40 cents, or \$4 per year. The annual net benefit to Bill is therefore \$1. The present value of the benefits = $\$5/0.25 = \20 ; the present value of the costs is $\$4/0.25 = \16 .

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d) The subway project has a higher present value. If a dollar to the “poor” is valued the same as a dollar to the “middle class,” choose the subway project.

e) Let λ be the distributional weight of a poor person relative to a rich person. The two projects are equivalent in terms of social welfare if

$$220,000 = -68,750 + \lambda \left(\frac{62,500}{1.25} + \frac{62,500}{[1.25]^2} \right)$$

which can be solved for $\lambda = 3.21$. In other words, this distribution weight means that \$1 of income to a poor person must be viewed as more important than \$3.21 to the middle class for the legal services to be implemented rather than the subway project.

Exercise 4.1

a) SW is additive and, hence, is maximal when $MU_A = MU_B$, which gives $\frac{\alpha}{I_A} = \frac{\beta}{I_B}$, or $I_B = \frac{\beta}{\alpha}I_A$. As $I_A + I_B = 400$, the optimal distribution is $I_A = 400\frac{\alpha}{\alpha+\beta}$ and $I_B = 400\frac{\beta}{\alpha+\beta}$. (Alternatively, you use Lagrange to maximise $U_A + U_B$ under constraint $I_A + I_B = 400$). As the SW-function gives the same weight to utility of oth individuals, the person who derives higher (marginal) utility from income, as given by α and β , optimally receives higher income.

b) Now, SW is maximal if $\omega_A MU_A = \omega_B MU_B$. This gives $\omega_A \frac{1}{I_A} = \omega_B \frac{1}{I_B}$, or $I_B = \frac{\omega_B}{\omega_A}I_A$. As $I_A + I_B = 400$, the optimal distribution is $I_A = 400\frac{\omega_A}{\omega_A+\omega_B}$ and $I_B = 400\frac{\omega_B}{\omega_A+\omega_B}$. (Alternatively, you use Lagrange to maximise $\omega_A U_A + \omega_B U_B$ under constraint $I_A + I_B = 400$). Now, utility functions are identical, but the SW function weighs utility differently (ω_A versus ω_B). The person who has a higher weight in the SW-function optimally receives more income.

Exercise 4.2

a) Social Welfare equals $W = \sqrt{I_D} + \sqrt{I_N}$, where $I_D + I_N = 2000$. Marginal utility of income equals $\frac{dU}{dI} = \frac{1}{2\sqrt{I}}$ for both. Given the additive SW function, welfare is maximised when marginal utilities are equal:

$$\frac{1}{2\sqrt{I_D}} = \frac{1}{2\sqrt{I_N}}$$

which yields $I_D = I_N$. As $I_D + I_N = 2000$, it must be that $I_D = I_N = 1000$ (Alternatively, you can substitute for $I_D = 2000 - I_N$ and maximise $W = \sqrt{2000 - I_N} + \sqrt{I_N}$).

b) Now we have that $I_D = 2000 - 1.1Y$ and $I_N = Y$. Substituting this into SW-function and maximising wrt Y gives

$$\begin{aligned} \frac{dW}{dY} &= -\frac{1.1}{2\sqrt{2000 - 1.1Y}} + \frac{1}{2\sqrt{Y}} = 0 \\ &\rightarrow \sqrt{2000 - 1.1Y} = 1.1\sqrt{Y} \end{aligned}$$

which can be solved for $Y \approx 865.80$. Hence, Nikki receives 865.80 and Dunya pays in taxes $1.1 \cdot 865.80 \approx 952.38$. Note that reduced efficiency of redistribution leads to less redistribution (and, hence, more inequality): trade-off between efficiency and distribution.

Exercise 4.3

a) Break-even point is the lowest amount of earnings E at which the individual does not receive any benefit. Benefit B is given by $B = G - tE$. Solving for $B = 0$ gives $E = G/t \rightarrow E = 1500$. Hence, after working 150 hours (or, equivalently, $L = 570$) she receives no benefit anymore.

b) Maximum earnings is 7200. Without welfare program, the budget constraint is $Y = 7200 - 10L$. Each hour of leisure has opportunity cost of 10 euro. With welfare program, the budget constraint consists of two parts: i) for $L \leq 570$: $Y = 7200 - 10L$; ii) for $L \geq 570$: $Y = 4920 - 6L$. The latter part we can derive from the fact that $Y = 600$ if $L = 720$ and $Y = 1500$ if $L = 570$.

c) This is maximisation with a constraint. Hence: either use Lagrange, or substitute the budget constraint into the utility function. The first method is probably best here. Lagrange equation is

$$\Gamma = Y^{\frac{1}{6}} L^{\frac{5}{6}} + \lambda (7200 - 10L - Y)$$

Partial differentiation wrt Y , L , and λ gives:

$$\begin{aligned} \frac{\partial \Gamma}{\partial Y} &= \frac{1}{6} Y^{-\frac{5}{6}} L^{\frac{5}{6}} - \lambda = 0 \\ \frac{\partial \Gamma}{\partial L} &= \frac{5}{6} Y^{\frac{1}{6}} L^{-\frac{1}{6}} - 10\lambda = 0 \\ \frac{\partial \Gamma}{\partial \lambda} &= 7200 - 10L - Y = 0 \end{aligned}$$

Combining the first and second gives $\frac{10}{6} Y^{-\frac{5}{6}} L^{\frac{5}{6}} = \frac{5}{6} Y^{\frac{1}{6}} L^{-\frac{1}{6}}$, or $2L = Y$. The last one gives $Y = 7200 - 10L$. Combining these two expressions gives $2L = 7200 - 10L$. Solving gives $L = 600$. Earnings are $10(720 - 600) = 1200$

d) As the optimal L without welfare is larger than 570, with welfare the second part of the budget constraint as derived under b) is the one Betty takes into account. Hence, the lagrange equation now equals

$$\Gamma = Y^{\frac{1}{6}} L^{\frac{5}{6}} + \lambda (4920 - 6L - Y)$$

Partial differentiation wrt Y , L , and λ gives:

$$\begin{aligned} \frac{\partial \Gamma}{\partial Y} &= \frac{1}{6} Y^{-\frac{5}{6}} L^{\frac{5}{6}} - \lambda = 0 \\ \frac{\partial \Gamma}{\partial L} &= \frac{5}{6} Y^{\frac{1}{6}} L^{-\frac{1}{6}} - 6\lambda = 0 \\ \frac{\partial \Gamma}{\partial \lambda} &= 4920 - 6L - Y = 0 \end{aligned}$$

Combining the first and second gives $Y^{-\frac{5}{6}} L^{\frac{5}{6}} = \frac{5}{6} Y^{\frac{1}{6}} L^{-\frac{1}{6}}$, or $L = \frac{5}{6} Y$. The last one gives $Y = 4920 - 6L$. Combining these two expressions gives $L = \frac{5}{6} [4920 - 6L]$. Solving gives $L = 683\frac{1}{3}$. Earnings equal $10(720 - 683\frac{1}{3}) = 366\frac{2}{3}$. Benefit equals $600 - 0.4 \cdot 366\frac{2}{3} = 453\frac{1}{3}$. Total income is 820. Utility equals 704.42. Hence, welfare makes

Betty better off (see e.g. Figure 13.4 in the book), but (in this example) her income is lower.

Some additional explanation on solving 4.3 c) and d):

$\frac{10}{6}Y^{-\frac{5}{6}}L^{\frac{5}{6}} = \frac{5}{6}Y^{\frac{1}{6}}L^{-\frac{1}{6}}$ can be written as

$$\begin{aligned}\frac{10}{6} \frac{L^{\frac{5}{6}}}{Y^{\frac{5}{6}}} &= \frac{5}{6} \frac{Y^{\frac{1}{6}}}{L^{\frac{1}{6}}} \\ 2 \frac{L^{\frac{5}{6}}}{Y^{\frac{5}{6}}} &= \frac{Y^{\frac{1}{6}}}{L^{\frac{1}{6}}}\end{aligned}$$

as $x^{-a} = \frac{1}{x^a}$. Next step is to multiply both sides with $Y^{\frac{5}{6}}L^{\frac{1}{6}}$. This gives

$$\begin{aligned}2 \frac{L^{\frac{5}{6}}Y^{\frac{5}{6}}L^{\frac{1}{6}}}{Y^{\frac{5}{6}}} &= \frac{Y^{\frac{1}{6}}Y^{\frac{5}{6}}L^{\frac{1}{6}}}{L^{\frac{1}{6}}} \\ 2L^{\frac{5}{6}}L^{\frac{1}{6}} &= Y^{\frac{1}{6}}Y^{\frac{5}{6}} \\ 2L &= Y\end{aligned}$$

as $x^a \cdot x^b = x^{a+b}$.

e and f) Now, $Y = 453\frac{1}{3} + 7200 - 10L$. Maximising Betty's utility yields the same relation between Y and L as under c): $2L = Y$. Hence, we get $12L = 7653\frac{1}{3} \rightarrow L = 637.78$ and $Y = 1275.56$. Utility equals $U = (1275.56)^{\frac{1}{6}}(637.78)^{\frac{5}{6}} = 715.88$. Betty works less than under c, as (given her utility function) the additional income implies that at the margin, leisure becomes relatively more attractive than the additional consumption generated by working (in other words: leisure is a normal good). She works more than under d), as the incentive to work is stronger (by working the same number of hours as under d, she would get the exact same utility. But now she gets an additional 10 euro income for working one more hour, rather than only 6 euro). She gets a higher utility than under d), implying that the welfare program under d) reduces efficiency (relative to a system that does not distort the choice between work and leisure as in e).

Exercise 4.4

a) SW is maximal if $MU_A = MU_B = MU_D$. This gives $\frac{1}{\sqrt{C_A}} = \frac{1}{\sqrt{C_B}} = \frac{w}{\sqrt{C_D}}$ or $C_A = C_B = C_D/w^2$. As $C_A + C_B + C_D = 2Y$ (total income is fixed), the optimal distribution is $C_D = \left(\frac{w^2}{2+w^2}\right) 2Y$ and $C_A = C_B = \left(\frac{1}{2+w^2}\right) 2Y$. Hence, $T = \left(\frac{w^2}{2+w^2}\right) Y$. (Alternatively, you use Lagrange to maximise SW under constraint $C_A + C_B + C_D = 2Y$).

b) An increase in w implies that marginal utility of income of Dirk increases. Through the (utilitarian) social welfare function, this makes giving more income to Dirk socially beneficial.

(c) Answer: Let T_A and T_B denote how much Anne and Bert give to Dirk, so that $C_A = Y - T_A$ and $C_B = Y - T_B$. Note that $C_D = T_A + T_B$. Substituting for C_A and C_D in U_A and maximizing with respect to T_A gives first-order condition

$$-\frac{1}{2\sqrt{Y - T_A}} + \frac{1}{4\sqrt{T_A + T_B}} = 0$$

which can be rewritten to $Y - T_A = 4(T_A + T_B) \rightarrow T_A = \frac{1}{5}(Y - 4T_B)$. Doing the same for Bert yields first-order condition

$$-\frac{1}{2\sqrt{Y - T_B}} + \frac{1}{4\sqrt{T_A + T_B}} = 0$$

which can be rewritten to $Y - T_B = 4(T_A + T_B) \rightarrow T_B = \frac{1}{5}(Y - 4T_A)$. (Beyond the course: the expressions for T_A and T_B are best-response functions).

We can find the equilibrium (= the levels of T_A and T_B such that given T_B , T_A maximises Anne's utility and given T_A , T_B maximises Bert's utility) by substituting for T_B in the expression for T_A . This yields $T_A = \frac{1}{5}(Y - 4(\frac{1}{5}(Y - 4T_A))) \rightarrow T_A = \frac{1}{25}Y + \frac{16}{25}T_A \rightarrow T_A = \frac{1}{9}Y$. Substituting this into the expression for T_B gives $T_B = \frac{1}{9}Y$. Beyond this question: this implies that $C_D = \frac{2}{9}Y$. For Anne and Bert, this is an inefficient outcome, as maximizing $U_A + U_B$ with respect to T_A and T_B would yield $C_D = T_A + T_B = \frac{2}{3}Y$.

d) Increasing T_A (and, hence, increasing U_D) has positive external effects on Bert, and vice versa. These positive external effects are ignored in deciding how much to increase U_D and, hence, there is underprovision of private poverty alleviation (= increasing C_D).

Alternatively, you can argue that from the perspective of Anna and Bert, helping Dirk is a public good (benefits are non-rival and non-excludable). Inefficiency arises because of free-rider behavior: both prefer the other person to pay for helping Dirk (more).

Exercise 4.5

a) True. Nora maximises $\sqrt{2500 - G} + \alpha\sqrt{400 + G}$ with respect to G , yielding first-order condition $-\frac{1}{2\sqrt{2500 - G}} + \alpha\frac{1}{2\sqrt{400 + G}} = 0 \rightarrow \alpha^2(2500 - G) = 400 + G \rightarrow G = \frac{2500\alpha^2 - 400}{1 + \alpha^2}$. Hence, $G > 0$ when $2500\alpha^2 - 400 > 0 \rightarrow \alpha^2 > 0.16 \rightarrow \alpha > 0.4$.

b) False. Aisha is better off with any gift $G > 0$. Nora is not worse off when $\sqrt{2500 - G} + \alpha\sqrt{400 + G} \geq \sqrt{2500} + \alpha\sqrt{400}$. Solving and substituting for $\alpha = 0.5$ and $G = 500$ yields $\sqrt{2000} + 0.5 \cdot 30 \geq 60 \rightarrow 59.72 \geq 60$ which is false.

c) True. Maximising $\sqrt{2500 - G} + 0.5\sqrt{600 + G}$ with respect to G yields first-order condition $-\frac{1}{2\sqrt{2500 - G}} + 0.5\frac{1}{2\sqrt{600 + G}} = 0$. Solving gives $0.25 \cdot (2500 - G) = 600 + G \rightarrow G = 20$.

d) True. For $\alpha > 0.4$, Nora would give a positive amount of money to Aisha (see a). This gift is reduced when Aisha receives more money (as Aisha's marginal utility of income reduces). See also lecture 4.